



## Solve with us 5 - Not Graded



This assignment will not be graded and is only for practice.

1. Key points:1. Key points:

- Rank of a matrix:
  - Maximum number of linearly independent column vectors (or row vectors) in a matrix.

NOTE: Rank of a matrix is always less than or equal to the number of columns (or rows).
- How to find rank of a matrix A:
  - Reduce the matrix to row echelon form.
  - Find the number of non zero rows in the reduced matrix which will be the rank of the matrix A.

1 point

Consider the following two matrices:

$$P = \begin{bmatrix} 1 & 0 & 3 \\ 2 & -2 & 0 \\ 0 & 1 & -1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 12 & 0 & 0 & -2 & 1 & 2 & 3 & 0 & -1 & 1 \end{bmatrix}; Q = \begin{bmatrix} 1 & 1 & -2 & 0 \\ 1 & 0 & 2 & 0 \\ 0 & -1 & 0 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & -1 & -2 & 2 & 0 & 0 & 4 \end{bmatrix}$$

Which of the following statements is (are) TRUE? [ Hint: Try to reduce the matrices to row echelon form and find the number of non zero rows ]

- ☐ Rank(P) = 4  
☐ Rank(P) = Rank(Q)  
☐ Rank(QP) = 4  
☐ Rank(PQ) = Rank(QP)

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

Rank(P) = Rank(Q)

Rank(PQ) = Rank(QP)

2 Key points:2 Key points:

- Null space of a matrix AA:
  - Let AA be an  $m \times n$  matrix. The subspace  $W = \{x \mid x \in R^n, Ax = 0\}$  is called the null space of AA.
  - The dimension of the null space W is called the nullity of AA.
- Rank-Nullity theorem:
  - Let AA be an  $m \times n$  matrix, then  $\text{rank}(AA) + \text{nullity}(AA) = n$ .

Consider the following matrix:



$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix} A = \begin{bmatrix} 0 & 0 & 0 & 1 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Use the matrix  $AA$  to answer the following questions (2) and (3).

1 point

Which of the following vector space represents the null space of  $A$ ?

☐

$\{(x_1, 0, x_2) \mid x_1, x_2 \in \mathbb{R}\}$   $\{(x_1, 0, x_2) \mid x_1, x_2 \in \mathbb{R}\}$

☐

$\{(0, x_1, 0) \mid x_1 \in \mathbb{R}\}$   $\{(0, x_1, 0) \mid x_1 \in \mathbb{R}\}$

☐

$\{(x_1, x_2, 0) \mid x_1, x_2 \in \mathbb{R}\}$   $\{(x_1, x_2, 0) \mid x_1, x_2 \in \mathbb{R}\}$

☐

$\{(0, 0, x_1) \mid x_1 \in \mathbb{R}\}$   $\{(0, 0, x_1) \mid x_1 \in \mathbb{R}\}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$\{(x_1, 0, x_2) \mid x_1, x_2 \in \mathbb{R}\}$   $\{(x_1, 0, x_2) \mid x_1, x_2 \in \mathbb{R}\}$

What will be the rank of the matrix  $AA$ ?

[Hint: Find the nullity of  $AA$  and use Rank-Nullity theorem]

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

3 Key points:3 Key points:

Linear transformation:

A function  $f$  from one vector space  $V$  to another vector space  $W$  ( $f: V \rightarrow W$ ) is said to be a linear transformation if for any two vectors  $v_1$  and  $v_2$  in the vector space  $V$  and for any  $c \in \mathbb{R}$  (scalar) the following conditions hold:

(i)  $f(v_1 + v_2) = f(v_1) + f(v_2)$

(ii)  $f(cv_1) = cf(v_1)$

1 point

In a super market, the rates of a pen, a water bottle, a tooth brush, and a chocolate are ₹ 10, ₹ 40, ₹ 25, and ₹ 20 respectively. Suppose  $T(x, y, z, w)$  denotes the total cost of  $x$  pens,  $y$  water bottles,  $z$  tooth brushes, and  $w$  chocolates. Which of the following will be the correct expression for  $T(x, y, z, w)$ ?

☐

$T(x, y, z, w) = x + y + z + w$

☐

$T(x, y, z, w) = 20x + 25y + 40z + 10w$

☐

$$T(x, y, z, w) = (x + y + z + w)(10 + 40 + 25 + 20) \quad T(x, y, z, w) = (x + y + z + w)(10 + 40 + 25 + 20)$$

☐

$$T(x, y, z, w) = 10x + 40y + 25z + 20w \quad T(x, y, z, w) = 10x + 40y + 25z + 20w$$

**No, the answer is incorrect.**

**Score: 0**

**Feedback:**

Check whether by evaluating for particular values of x, y, z, w the function value matches the cost.

**Accepted Answers:**

$$T(x, y, z, w) = 10x + 40y + 25z + 20w \quad T(x, y, z, w) = 10x + 40y + 25z + 20w$$

*1 point*

Suppose Keerthana bought 3 pens, 5 water bottles, 10 chocolates and she did not buy any tooth brushes.

Which of the following options denotes the total cost paid by her?

☐

$$T(3, 5, 1, 10) \quad T(3, 5, 1, 10)$$

☐

$$T(3, 5, 0, 10) \quad T(3, 5, 0, 10)$$

☐ ₹430

☐ ₹450

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$T(3, 5, 0, 10) \quad T(3, 5, 0, 10)$$

₹430

[Check Answers](#)

Your score is: 0/5